

Before you start working on the project, you need to spend some time on the “**Problem Definition**”.

As you recall from reading Chapter 3 in Fogler and LeBlanc, **Problem Definition** has several steps and processes that can be used. You are expected to do three things: gather information, define the problem using several of the problem definition techniques, and finally evaluate the problem definition you eventually come up with.

The Assignment:

Your assignment is to undertake the problem definition task that **had not been previously done**.

The Situation: Students do not know enough about manufacturing

(recommended) functional solution: Develop a manufacturing project for the first year engineering course

1. **Step : Information Gathering**

The suggested, first, four steps in Problem Definition (see Table 3-1, page 29 in Fogler and LeBlanc) involve gathering information. You are going to want to do at least steps 1 and 4 from Table 3-1.

2. **Step 2 Problem Definition**

Now that you have gathered your information, it is time to work on getting a good problem definition (*minimum get **THREE** different definitions*). It is expected that you use **the Present State/Desired State** process, the **Statement/Restatement** Process, and at least one Duncker Diagram as you try to work out a good problem definition. The **recommended-functional-solution** stated above must be among the functional solutions.

3. **Step 3 Evaluation of the generated Problem definitions**

Section 3.2F (page 50) lists a series of questions (shown as bullets) you can ask and answer as you evaluate the problem definition you have generated.

Remember goal of problem definition work should be to **DEFINE** the real problem

NOT to do the work necessary to solve the problem.

The Work in the DNB:

- 1- This assignment is a series of tasks that, if taken together, constitute a **Problem definition** effort.
- 2- Your work should be developed and presented according to the class expectations (PTW)
- 3- We (the Laboratory Instructors, your customers) are **NOT LOOKING** for a report; rather we are looking in your DNB for an **organized presentation** of your work.
- 4- **The work need not be word-processed** but it must be neat and readable. Word processing everything is not a way to exceed expectations.
- 5- It is expected that at least part of your discussion will be a **reflection back** on the process that was undertaken in doing the completed task.
- 6- Most of this work will be done as a team during team meeting(s). It is important to not lose the discussion that takes place as the team carries out various exercises.
- 7- It is expected that there will be more than just the final result (e.g., Present State/Desired State, Duncker diagram) in the notebook; **the important parts of the team discussion that lead to the values in the work are expected to also be present**.
- 8- Don't forget to include evidence that you are functioning as a team (team meeting agendas/minutes, etc.)
- 9- Without losing generality: the main DNB-sections (not in order) are: the assessment section, the project section (including L4 (a-e)), the lab-assignment section (including L1, L2 and L5), Quizzes, meeting minutes, Lecture process check, miscellaneous section etc.

The Assessment:

The work will be self-assessed according to the 1st DNB assessment checklist items. Peer assessment will be done in class under the supervision of the instructor. The instructor will do the final assessment.

Now that you understand the project; it is time to generate a rich set of possible artifact names and set of constraints (**musts**) and desires (**wants**) that can be used in helping your team select a great artifact.

The Assignment: Your specific assignment is as follows:

1) Task 1: What are the Possible Artifacts?

- a) Generate a large set of possible **artifacts** that the team could use. It is expected that the set will be at least **20** items long.
- b) Use multi-voting (see page K-15 in the Blue Workbook) to reduce the number of possible artifacts down to **between six and ten**.

2) Task 2: What are the Constraints?

- a) Generate a list of at least **10** possible constraints (**musts**, i.e., things that must happen or be true). For example, you must be able to obtain the artifact that the team selects. The course instructor does not want you to spend much money on an artifact, however, it is up to the team. There are, of course, other ways besides buying a new item to obtain a desired artifact (used items, or something donated). It is expected that you will use Osborn's Checklist to increase your set of constraints as soon as "brain drizzle" sets in.
- b) Review each constraint and assess whether the constraint is real or just the result of brainstorming; remove any constraints that are not needed. Use multi-voting to reduce the list of musts down to **three** (at least).

3) Task 3: What are your Desires?

- a) Generate at least **10** possible team desires (**wants**, i.e., things the team would like to be true or happen). For example, the team might want to have many parts when the artifact is dissected. It is expected that you will use Random Stimulation to increase your set of desires at the point "brain drizzle" sets in.
- b) use multi-voting to reduce the list of desires down to **three or four**.

4) Task 4: Summarize: Summarize the final selection out of task 1, 2 and 3 in the given summary sheet by hand writing?

The Work In The Notebook

This assignment is a series of tasks that, if taken together, constitute the Idea Generation effort. Your work should be developed and presented according the class expectations concerning presentation of technical work (i.e., there should be a context and discussion for the assignment). We (the Lab Instructors, your customers) are **NOT LOOKING** for a report; rather we are looking in your DNB for an organized presentation of your work. **The work need not be word processed** but it must be neat and readable. Word processing everything is not a way to exceed expectations.

Much of this Idea Generation work will tend to be "messy"; idea generation is not a simple, logical, linear task. With some "effort" you will be able to use your work directly in your "sandwich" but if the work is too messy and really makes little sense, you may have to clean it up a bit and add a few comments about what was going on.

Don't forget to include evidence that you are functioning as a team (meeting agendas, minutes, etc.)

The Assessment

The work will be self-assessed according to the 2nd DNB assessment checklist items. Peer assessment will be done in class under the supervision of the instructor. The instructor will do the final assessment.

SUMMARY OF L-4-b	TEAM: Use the back if required
<i>This sheet must be completed in handwriting before submission</i> <i>Keep copy for the team use before submitting it</i> <i>This sheet must be located as an appendix to the associated assignment</i>	

Sem

1- **ARTIFACT names (in English)**

1	
2	
3	
4	
5	
6	

2- **MUSTS (in English)**

1	
2	
3	
4	
5	

3- **WANTS (in English)**

	The want	Its weight
1		
2		
3		
4		
5		
6		

The 1st DNB Assessment Checklist (2026) - PI-1.3

Team _____

Work Products Being Evaluated: **Project (Assignment #L-4-a)**

1. Work Evaluated by (**self**) _____ Assessment Symbol/Color _____ Date _____
2. Work Evaluated by (**peer**) _____ Assessment Symbol/Color _____ Date _____
3. Work Evaluated by (**instructor**) _____ Assessment Symbol/Color _____ Date _____

Yes	No	Expected Features	Re
		1. This Checklist is completed (self-assessed) before submission and is in the assessment section	
		2. The TOC has been UPDATED to reflect the new work (page numbers are shown for all tasks/sub tasks)	
		3. The context for project L-4 is located in a subsection within the project section	
		4. The PTW is used as the cover sheet of L-4-a's assignment, -as one set- located at the appropriate section in the DNB and was assessed as "M", <i>without any other coversheet.</i>	
		5. L-4-a: there is work for Exploring the Problem	
		6. L-4-a: contains a correct usage of Statement/Restatement	
		7. L-4-a: contains a correct usage of Present State/Desired State (includes analysis)	
		8. L-4-a: contains a correct Duncker Diagram , right hand side is solving a problem.	
		9. L-4-a: There is at least one specific solution for each functional solution in DD	
		10. L-4-a : it is clear which functional/specific solution is selected in Duncker Diagram	
		11. L-4-a: has at least Three different problem definitions	
		12. L-4-a: contains EVALUATION for each of the problem definitions.	
		13. It is clear what is the selected/preferred problem definition among the furnished definition	
		14. According to the selected problem definition, is the problem open-ended?	
		15. At least 9 concept -lecture-summary are located in the appropriate section. <i>Any work done inside classes must be included (like Active learning, dropping the tomato ...) (staple each lecture separately)</i>	
		16. At least 9 Lab -lecture-summary are located in the appropriate section. <i>Any work done inside classes must be included (like Air volume, flowchart) (staple each lecture separately)</i>	
		17. The lecture-process-checks (whatever completed TILL NOW) are in a separate section.	
		18. The Quiz-section has all the original quizzes performed and returned to students till now	
		19. There is a section called assignments (suitable to add lab assignments, other than the project, as L1, L2, L5)	
		20. The team photo is located (glued) on the back of the left cover	
		21. The discussion of assignment L-3 is updated and located by the end of the DNB	
		22. There is a miscellaneous section that contains what is not enough to open separate section; <i>If there is no such section type 'NA' in the yes cell</i>	
		23. In general; the folder of the DNB is solid, strong, cohesive, in good shape and all included papers are placed with the correct orientation	

A specific cell in the most right column is to be checked in the re-submission only 

Comments on any of the No's

Wow	Ok	Weak	Revealed Features
			Rate the quality of the notebook organization (e.g., aesthetics, use of unique page numbers, current TOC, reference to other work in the notebook, etc.)

The 1st DNB Assessment Checklist (2026) - PI-1.3

			Rate the <u>initial** quality & completeness of the context of L-4</u>
			Rate the <u>initial** quality & completeness of the work done in exploring the Problem</u>
			Rate the quality & completeness of <u>the work done to generate the Problem definitions</u>
			Rate the quality and completeness of <u>the evaluation of the problem definitions</u>
			Rate the <u>initial**</u> quality of the <u>Discussion</u> of L-4-a (e.g., how good was the reflection, aesthetics)
			Rate the <u>readability of the work</u> (i.e., the level of communication "noise" which consists of grammatical, spelling, syntax, etc. errors that hinder communication)
Yes	Exciting Feature(s)		

Results of Initial Assessment

*Initial means**: this evaluation may differ from the evaluation of the same work by the end of the semesters. By the end of the semester, the view and outcomes would be clearer than now.*

Total Number of WOW :		Total Number of WEAK :	Total Number of NO :
If needed; Instructor may conclude (1-WOW and 1-WEAK) to two Oks			
If needed; Instructor may conclude (1-WOW and 1-NO) to one Ok & 0 No (this option is added to this checklist only!)			
Final Number of WOW :		Final Number of WEAK :	Final Number of NO :
E	- At least two WOWs or (One WOW and an <u>Exciting Feature</u>), and - Zero weak, zero NO, Zero lapse		
M	All <u>Yes</u> for the Expected Features – one weak only is accepted for (M)		
NI	No more than three as the total sum of weaks and Nos		
NCE	Any other situations		
NS	The folder is not rigid enough to be handled by the grader or was not submitted		

*If the work was assessed as **NI**, **NCE** or **NS** you may, following the information given in the LAB-FDM and the instructions given on the Resubmission of Work Checklist (if **applicable**), improve the work and resubmit it (if **applicable**) for re-assessment.*

----- students do not write after this line -----

For instructor use only: check the box to the right if it was required (from this team) to upload the word-processed part of this assignment to the BB. The upload must be before C-5-3 to the percentage of the plagiarism	
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The **2nd** DNB Assessment Checklist **2026** -PI-2.1

Team _____

Work Products Being Evaluated: **Project (Assignment #L-4-b)**

1. Work Evaluated by (self) _____ Assessment Symbol/Color _____ Date _____
2. Work Evaluated by (peer) _____ Assessment Symbol/Color _____ Date _____
3. Work Evaluated by (instructor) _____ Assessment Symbol/Color _____ Date _____

Yes	No	Expected Features	Re
		1. This Checklist is completed (self-assessed) before submission and is located in the assessment section	
		2. The TOC has been UPDATED to reflect the new work (page numbers are shown for all tasks/sub tasks)	
		3. The PTW is used as the cover sheet of L-4-b 's assignment and was assessed as "M", without any other coversheet.	
		4. INITIALLY ; in L-4-b , there were at least 20 artifacts , 10 constraints (musts), and 10 desires (wants) from each member <i>the original team may state the evidence here</i>	
		5. There is a clear final list , in L-4-b , of at least 6-artifacts , 3-musts and 3 desires	
		6. There is explicit evidence that the team used correctly multi-voting (<i>for artifacts, musts and wants</i>) – <i>the original team may state the evidence here</i>	
		7. There is explicit evidence that the team used Osborn's Checklist (<i>for artifacts, musts and wants</i>) <i>the original team may state the evidence here</i>	
		8. There is explicit evidence that the team used Random Stimulation (<i>for artifacts, musts and wants</i>) <i>the original team may state the evidence here</i>	
		9. The summary sheet of the final ideas (Artifacts, Musts and wants) is located as one of the appendices of the L-4-b's assignment) and is completed by hand writing	
		10. The discussion of assignment L-3 is updated and located by the end of the DNB	
		11. There is a miscellaneous section that contains what is not enough to open separate section; <i>If there is no such section type 'NA' in the yes cell</i>	
		12. In general; the folder of the DNB is solid, strong, cohesive, in good shape and all included papers are placed with the correct orientation	
A specific cell in the most right column is to be checked in the re-submission only			
<u>Comments on Expected Features and how any problems might be resolved</u>			

Wow	Ok	Weak	Revealed Features
			1. Rate the quality of the context of the work
			2. Rate the readability of the work (i.e., the level of communication "noise" which consists of grammatical, spelling, syntax, etc. errors that hinder communication)

The **2nd** DNB Assessment Checklist **2026** -PI-2.1

Wow	Ok	Weak	Revealed Features
			3. Rate the quality of the notebook organization (e.g., use of unique page numbers, reference to other work in the notebook, etc.)
			4. Rate the quality of the initial** set of constraints generated (the set of constraints is neither too restrictive nor too permissive)
			5. Rate the quality of the initial** set of wants generated (the set of unmeasurable wants are not required)
			6. Rate the quality of the Discussion (e.g., how good was the reflection)
			7. In general, rate the effort paid to perform L-4-b
Yes	Exciting Features		

Results of Initial Assessment

*Initial** means: this evaluation may differ from the evaluation of the same work by the end of the semesters. By the end of the semester, the view and outcomes would be clearer than now.*

Total Number of WOW :		Total Number of WEAK :	Total Number of NO :.....
E	- At least two WOWs or (One WOW and an <u>Exciting Feature</u>), and - Zero weak, zero NO, Zero lapse		
M	All <u>Yes</u> for the Expected Features – one weak only is accepted for (M)		
NI	No more than three as the total sum of weaks and Nos		
NCE	Any other combination		
NS	The folder is not rigid enough to be handled by the grader or was not submitted		
It is not permitted, in this work, to conclude WOWs with NOs and/or WEAKs			

----- Students: do not write after this line -----

For instructor use only: check the box to the right if it was required (from this team) to upload the word-processed part of this assignment to the BB. The upload must be before C-5-3 to the percentage of the plagiarism	
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*If the work was assessed as **NI**, **NCE** or **NS** you may, following the information given in the LAB-FDM and the instructions given on the Resubmission of Work Checklist (if applicable), improve the work and resubmit it (if applicable) for re-assessment.*